

Probability and Independence

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Overview

Independence (17.6)

Alternative Formulation (17.6.1)

Mutual Independence (17.6.3)

Conditional probability

Definition: The conditional probability of event A given event B is:

$$\Pr[A|B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

Conceptually, if we limit ourselves to the outcomes in B , how likely is an outcome in A ?

Example:

- ▶ A : Die shows a number divisible by 3. $\Pr[A] = 1/3$. (3 and 6 from the six possibilities.)
- ▶ B : Die shows an odd number. $\Pr[B] = 1/2$.
- ▶ What does $\Pr[A \cap B]$ mean? Die shows an odd number divisible by 3. $\Pr[A \cap B] = 1/6$ (only 3).
- ▶ What does $\Pr[A|B]$ mean? Die shows a number divisible by 3 given that it's odd. $\Pr[A|B] = 1/3$ (probability of picking 3 from 1, 3, 5). Also, $\frac{1}{6} / \frac{1}{2} = \frac{1}{6} \times 2 = \frac{1}{3}$.

Independence

Definition: Event A is *independent* of event B iff

$$\Pr[A|B] = \Pr[A].$$

If $\Pr[B] = 0$, we say it is independent of any other event including itself.

Example:

- ▶ A : Die shows the maximum or minimum number. $\Pr[A] = 1/3$. (1, 6 from the six possibilities.)
- ▶ B : Die shows an odd number. $\Pr[B] = 1/2$.
- ▶ $\Pr[A|B] = 1/3$. (1 from 1,3,5.) So, A and B are independent.
- ▶ C : Die shows an even number. $\Pr[C] = 1/2$.
- ▶ Are B and C independent? No, $\Pr[C|B] = 0 \neq \Pr[C]$.
Common misconception. Independent does not mean disjoint.

Independent events multiply

Theorem: A is independent of B iff

$$\Pr[A \cap B] = \Pr[A] \cdot \Pr[B].$$

Proof: By cases.

- ▶ Case 1: If $\Pr[A] = 0$ or $\Pr[B] = 0$, then $\Pr[A \cap B] = 0$. Equality and independence are both achieved.
- ▶ Case 2: Otherwise, A is independent of B iff $\Pr[A|B] = \Pr[A]$ by definition. Substituting in the definition of conditional probability, we have $\Pr[A|B] = \Pr[A]$ iff $\Pr[A \cap B]/\Pr[B] = \Pr[A]$. Multiplying both sides by $\Pr[B]$, we have $\Pr[A \cap B] = \Pr[A] \cdot \Pr[B]$. QED.

Mutual Independence

Definition: A set of events $E_1, E_2, \dots, E_n$ is *mutually independent* iff for all subsets $S \subseteq [1, n]$,

$$\Pr \left[\bigcap_{j \in S} E_j \right] = \prod_{j \in S} \Pr[E_j].$$

Example: If we toss n fair coins, the tosses are mutually independent iff for every subset of m coins, the probability that every coin in the subset comes up heads is 2^{-m} .

Independent missions

Mars	one-time	person	.028
Mars	one-time	robot	.042
Mars	reusable	person	.252
Mars	reusable	robot	.378
Moon	one-time	person	.012
Moon	one-time	robot.	.018
Moon	reusable	person	.108
Moon	reusable	robot.	.162

$$\Pr[\text{Mars}] = 0.028 + 0.042 + 0.252 + 0.378 = 0.7$$

$$\Pr[\text{one-time}] = 0.028 + 0.042 + 0.012 + 0.018 = 0.1$$

$$\Pr[\text{person}] = 0.028 + 0.252 + 0.012 + 0.108 = 0.4$$

$$\Pr[\text{Mars AND one-time}] = 0.028 + 0.042 = 0.07 = 0.7 \times 0.1$$

$$\Pr[\text{Mars AND person}] = 0.028 + 0.252 = 0.28 = 0.7 \times 0.4$$

$$\Pr[\text{one-time AND person}] = 0.028 + 0.012 = 0.04 = 0.1 \times 0.4$$

$$\Pr[\text{Mars AND one-time AND person}] = 0.028 = 0.7 \times 0.1 \times 0.4$$

Pairwise independence isn't mutual independence

It is natural to think pairwise independence implies mutual independence. If a is independent of b and c , and b and c are independent of each other, how could a , b , and c not be independent??

Example: Morley, Kyran, and Will each pick a bit. If I tell you the sum of of Morley and Kyran's bits (mod 2), do you learn anything about either of their bits? No.

But, what if I tell you two of these pairwise sums mod 2?

Morley + Kyran 1

Morley + Will 0

Will + Kyran 1

Must be 1 because we know Morley doesn't match Kyran, but Morley and Will match. So, Will and Kyran can't match.

k -wise does not imply $(k + 1)$ -wise mutual independence.